EXECUTIVE SUMMARY

Size gold risk on ranges adjusted for the time of day, not on forecasts of where the price will go

95.8%

of hourly prices fell inside the 95% range over six unseen weeks

4×

gap in volatility between the busiest hour (8 a.m. New York) and the quietest (4 p.m.)

0 of 3

price models, including an LSTM neural network, that beat "the price stays the same"

The question

How far could the price of gold in Hong Kong dollars move in the next hour, week or month, and can that answer be trusted?

The evidence

185,071 minutes of real quotes, cleaned into 3,090 hourly prices. Ranges were tested over six rolling weeks (687 hours) and on the first week of November, which no model saw.

The answer

- The direction of the next move could not be predicted from past prices.
- The size of moves could: it clusters, and follows the clock.
- A volatility model adjusted for the time of day hit its target on unseen weeks, but was too narrow in November's sharp rally.

THE SITUATION

Risk limits and hedges are sized on how far gold could move, so that range has to hold on new data

3,090
hourly prices of gold in Hong Kong dollars, May to November 2026

7 weeks
of testing on data the models never saw: six rolling weeks, then early November

5
problems fixed before trusting the results, such as a one-hour shift when US clocks changed

Why it matters

Anyone holding, hedging or pricing gold in Hong Kong dollars needs to know how far the price could move, not just where it might go. A range that is too narrow understates risk, and one that is too wide ties up capital.

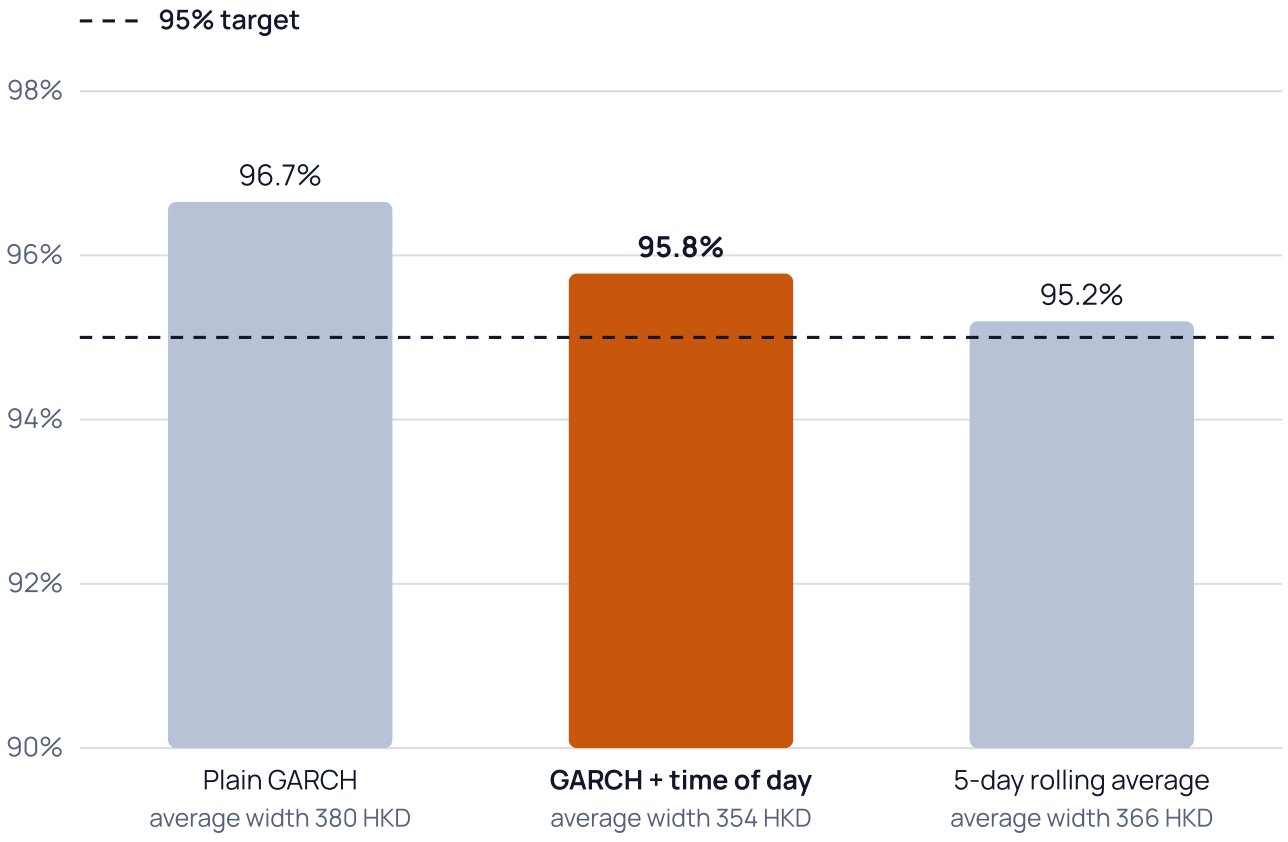
What was done

1. Forecast volatility with GARCH, a model of how much prices are likely to move, adjusted for the time of day and weekend gaps.
2. Refitted it every week, the way a risk team would use it, and checked how often prices fell outside.
3. Tested whether an LSTM, ARIMA or a trend model could predict the price itself.

FINDING 1

Adjusting for the time of day hit the 95% target with the narrowest range

Share of hours inside each model's 95% range, six unseen weeks



- **On target.** 95.8% of hours fell inside, and a standard test found the miss rate in line with 95% ($p = 0.34$). The misses were spread out, not bunched.
- **Plain GARCH was too cautious.** 96.7% inside, with a wider range than needed.
- **A rolling average missed in clusters.** Its misses bunched together ($p = 0.02$), so it fails exactly when risk is high.

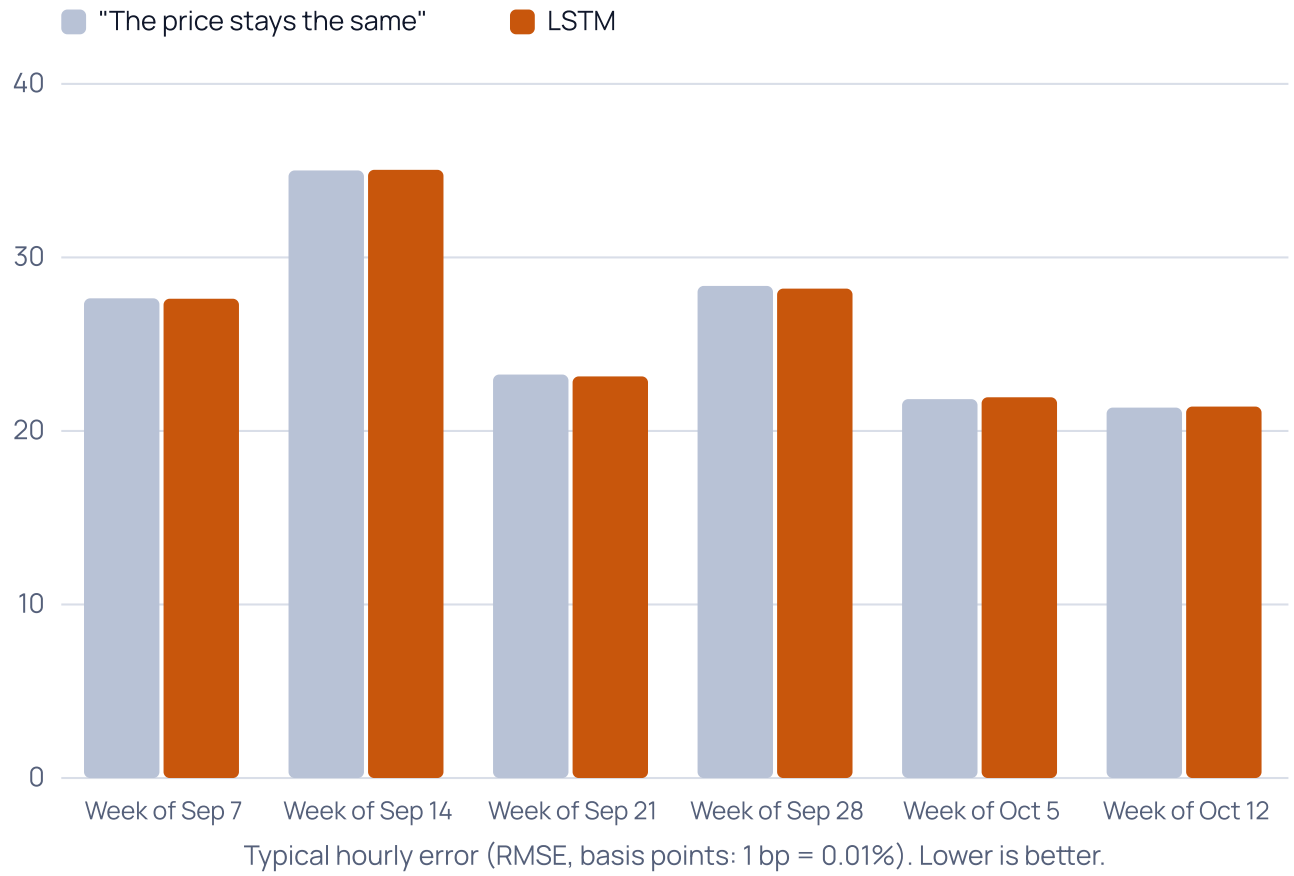
SO WHAT

Volatility at 8 a.m. New York time, when US economic data come out, is about four times that at 4 p.m. Building this in gives tighter ranges at no cost to coverage.

FINDING 2

No model could predict the price, so ranges are centred on the latest price

Typical hourly forecast error, week by week



Typical hourly error (RMSE, basis points: 1 bp = 0.01%). Lower is better.

- **A coin toss.** Over 687 unseen hours the LSTM's error was within 0.05% of "the price stays the same" ($p = 0.77$), and it was ahead in only 3 of 6 weeks.
- **Memorising noise.** Trained for longer, the LSTM got better on past data and worse on new data.
- **Same for simpler models.** ARIMA and a trend model did no better.

> SO WHAT
>
> A price forecast would add false confidence. The value is in forecasting how far the price could move.

RECOMMENDATION

Three moves for a risk team, with one honest miss to fix first

What to do

1. **Use the time-of-day ranges for hourly limits**
   Refit them every week, as in testing.
2. **Add a daily volatility model for week-ahead ranges**
   So longer ranges react faster to a rally.
3. **Do not trade on price forecasts**
   None beat "the price stays the same".

Next step

Check the 80% range set for 30 November, 29,975 to 34,241 HKD per ounce, once the month ends.

Limits to keep in mind

- **November was a miss.** Gold rose 5.3% in four days. The 1-hour 95% range held 89.2% of hours, below target, and the week-ahead range was too narrow.
- **Six months of data**, so calmer or more turbulent periods may behave differently.
- **One price.** Only the buying (ask) price of gold in Hong Kong dollars was modelled.
- **Not yet tried.** No risk team has used these ranges.